

SignTalk AI — Product Requirements Document (PRD)

Product requirements, user needs, functional scope and acceptance criteria

SignTalk AI | B.Tech CSE (AI/ML) | GLA University

1. Product Vision

Create an accessible communication layer that helps supported ISL signing users communicate with non-signers by converting continuous signing into readable text with low latency and privacy-aware processing.

2. Target Users

Persona	Need	Outcome
Deaf / hard-of-hearing signer	Communicate with non-signers	Real-time readable text
Healthcare/public-service user	Rapid communication	Fast captions
Non-signer	Understand signing	Clear sentence output
Organization/integrator	Add translation to a platform	Future API/SDK

3. Primary User Journey

1. Open application.
2. Grant camera permission.
3. Select ISL and output language where supported.
4. Start translation.
5. Perform supported signs continuously.
6. System extracts landmarks.
7. AI interprets temporal sequence.
8. Text appears as live captions.
9. Repeat unclear content when confidence is low.
10. End session.

4. Functional Requirements

ID	Requirement	Priority	Acceptance
FR-01	Live camera input	Must	Continuous frames processed
FR-02	Hand landmarks	Must	Finger/hand landmarks available
FR-03	Body landmarks	Must	Relevant pose keypoints available
FR-04	Facial/non-manual features	Should	Features available where enabled
FR-05	Normalize/fuse features	Must	Unified representation produced
FR-06	Continuous supported-sign recognition	Must	Supported sequence produces prediction
FR-07	Text generation	Must	Sequence produces

			readable text
FR-08	Live captions	Must	Caption updates continuously
FR-09	Confidence indication	Should	Low-confidence output visibly flagged
FR-10	Offline core inference	Must/Target	Works without network on supported hardware
FR-11	Language selection	Could	Supported options exposed
FR-12	API/SDK	Could	Future external clients can consume output

5. Non-Functional Requirements

Area	Requirement
Performance	Target end-to-end latency <500 ms; measure actual result
Privacy	No mandatory raw-video cloud upload
Accessibility	Readable captions, high contrast, keyboard access
Usability	Minimal steps to start translation
Scalability	Modular inference/API boundary
Maintainability	Version code, data and models
Observability	Measure latency, FPS, errors and model metrics
Security	Protect sessions/secrets and avoid sensitive logs

6. UX Requirements

Main screen: camera preview, large high-contrast live caption area, confidence indicator, language selector, Start/Stop controls and privacy/offline status.

- Minimal controls
- Large readable captions
- Clear camera status
- Low cognitive load
- Visible uncertainty
- No unnecessary data collection
- Keyboard accessibility

7. Error States

Condition	Expected behavior
Camera unavailable	Clear permission/device error
No hand detected	Do not invent a sign; show detection message
Low confidence	Flag result and prompt repeat
Unsupported sign	Show unknown/unsupported state
Poor lighting	Warn when landmark quality degrades
No network	Continue local inference if supported
Model failure	Show safe temporary error
Multiple signers	Prototype defines a primary-signer assumption

8. Success Metrics

Goal	Metric
Recognition	Accuracy / Precision / Recall / F1
Translation	BLEU/ROUGE/semantic similarity
Real time	End-to-end latency
Responsiveness	FPS
Robustness	Performance by condition
Usability	Task completion/user feedback
Privacy	Observed video transmission behavior

9. Release Plan

Release	Scope
Prototype	Camera + landmarks + baseline recognition
Model release	ST-GCN + sequence translation
Alpha	Backend + real-time captions
Optimization	ONNX/quantization experiments
Academic v1.0	Validated demo + evaluation + documentation

10. Acceptance Criteria

- Camera can start/stop reliably.
- Supported sequences are evaluated on held-out data.
- Continuous supported signing produces captions.
- Uncertain input is explicitly flagged.
- Offline inference works on selected hardware if included.
- Dataset split and test conditions are documented.
- Raw video is not retained by default.

11. Out of Scope

- Universal ISL coverage
- Automatic legal/medical interpretation
- Guaranteed perfect translation
- Native mobile app initially
- Large-scale cloud video storage
- Automatic real-world actions from translated content